Unit 2.1 Printables

DNA to Proteins (HS-LS1-1) · 8-day version: Siamese Cats

Everything students write goes in the notebook (Unit_2.1_Student_Notebook.pdf). These pages are only the extras.

Day	Print	Copies
1, 3, 4, 7	Glue-in strips (page 2): Day 1 key ideas, Day 3 base key and code table, Day 4 code table, Day 7 Write It data	1 per student (cut apart)
3	Gummy DNA mats (page 3)	1 per pair
4	Card sort, cards A to F (page 4)	1 set per group

Day 3 lab materials (per pair)

- 2 licorice strands (6 to 8 inches)
- 24 candy bases: 6 red (A), 6 yellow (T), 6 green (C), 6 orange (G), plus a few spare
- Toothpicks, a paper plate
- No food? Use colored beads and pipe cleaners, or colored paper circles

Cut on the dashed lines. Strips are sized to fit half of a notebook page.

Glue-in strips

Cut apart on the dashed lines and glue into the notebook.

Day 1 · Part 2: Key ideas

- **DNA** in every cell stores instructions.
- A **gene** is a section of DNA with the instructions for one **protein**.
- **Proteins** do jobs for the cell.
- A kitten gets one set of chromosomes (long DNA molecules with many genes) from each parent.

Day 3 · glue in above Part 1: Lab record

Red = A Yellow = T Green = C Orange = G

DNA triplet	Amino acid	DNA triplet	Amino acid
ATG	Met	AGA	Arg
GGA	Gly	CTG	Leu
TTT	Phe	GAC	Asp

Day 4 · glue in at Part 2: Card sort

DNA triplet	Amino acid	DNA triplet	Amino acid
GGA	Gly	AGA	Arg
GAG	Glu	GTG	Val
GAA	Glu	CTG	Leu
CCG	Pro		

Day 7 · Part 3: Write It data, two Siamese kittens

	Kitten raised warm	Kitten raised cool
Ear skin temperature (°C)	34	26
Ear fur color	Light tan	Dark brown
Back skin temperature (°C)	36	35
Back fur color	Cream	Cream

Day 3 · Gummy DNA mats

Slide the licorice strand along the mat. One base per box, three boxes per amino acid.

Parts A and B: template strand (T A C C C T G A C A A A) and partner strand

Template



Partner



Part C: the copy



Amino acids: _____ – _____ – _____ – _____

Part D: changed template (T A C T C T G A C A A A) and its copy





Amino acids: _____ – _____ – _____ – _____

Day 4 · Card sort

Cut on the dashed lines. Sort into three categories, glue into the notebook, and write your reasons.

A

In a Siamese cat, one letter changes in the gene for a pigment-making protein: GGA becomes AGA.

B

A gene's triplet changes from GAG to GAA.

C

A gene's triplet changes from CTG to CCG.

D

One letter changes in a stretch of DNA that has no gene and no switch.

E

One letter changes in a switch next to a gene so the gene can never be turned on, and its protein is not made.

F

In a human gene, GAG becomes GTG.